

Sanghyeon Lee

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. in Artificial Intelligence

Kim Jaechul Graduate School of AI

September 2021 – Present

Advisor: Prof. Jaegul Choo

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Artificial Intelligence

Kim Jaechul Graduate School of AI

September 2019 – February 2021

Advisor: Prof. Jaegul Choo

Seoul National University

B.S. in Economics

Completed mandatory military service, 2013–2015

March 2011 – August 2018

RESEARCH PROFILE

I am interested in agentic visual AI systems that understand human intent, plan controllable generation and editing operations, and eventually realize visual outputs in the physical world through embodied models.

RESEARCH INTERESTS

Agentic Human-AI Interaction for Visual Creation; Controllable Image Generation; Embodied Visual Creation with Vision-Language-Action Models.

TEACHING INTERESTS

Machine Learning and Deep Learning; Computer Vision; Generative Models; Human-AI Interaction; Data Science; Python Programming. Mathematical Statistics

PEER-REVIEWED CONFERENCE PUBLICATIONS

*Equal contribution.

- [C5] **Sanghyeon Lee**, Minwoo Lee, Euijin Shin, Kangyeol Kim, Seunghwan Choi, and Jaegul Choo. OPRO: Orthogonal Panel-Relative Operators for Panel-Aware In-Context Image Generation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. Accepted (25.4% acceptance rate). **CVPR Compute Transparency Champion**. [paper]
- [C4] **Sanghyeon Lee**, Jooyeol Yun, and Jaegul Choo. Enabling Region-Specific Control via Lassos in Point-Based Colorization. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025. Accepted (23.4% acceptance rate). [paper] [code]
- [C3] Jooyeol Yun*, **Sanghyeon Lee***, Minho Park*, and Jaegul Choo. iColoriT: Towards Propagating Local Hints to the Right Region in Interactive Colorization by Leveraging Vision Transformer. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023. Round 1 accept (22% Round-1 acceptance rate). [paper] [project page]
- [C2] Eungyeup Kim*, **Sanghyeon Lee***, Jeonghoon Park*, Somi Choi, Choonghyun Seo, and Jaegul Choo. Deep Edge-Aware Interactive Colorization against Color-Bleeding Effects. *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. **Oral presentation** (top 3.4% of submissions). [paper] [project page]
- [C1] Youngwoo Cho*, Minwook Chang*, **Sanghyeon Lee**, Hyungwoo Lee, Gerard Jounghyun Kim, and Jaegul Choo. Efficient Adversarial Audio Synthesis via Progressive Upsampling. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021. Accepted (48.0% acceptance rate). [paper]

MANUSCRIPTS UNDER REVIEW

- [U1] **Sanghyeon Lee***, Mina Hwang*, Soyoung Lee, Jaegul Choo, and YoungJoon Yoo. Iterative Coarse-to-Fine Refinement for Virtual Try-On with VLM-Guided Region Selection. Under review.

RESEARCH PROJECTS

National Institute of Meteorological Sciences Research Project	South Korea
Researcher	2025 – Present
– Developing a tool-augmented agentic LLM for weather forecasting support, including domain-specific tool design, dataset construction, and model training.	
Electronics and Telecommunications Research Institute (ETRI) Research Project	South Korea
Researcher	2022 – 2023
– Developed a sketch-to-image colorization model using point- and reference-based inputs for interactive image generation.	
Samsung Electronics Research Project	South Korea
Researcher	2021
– Developed a reference-based photo enhancement model.	

PROFESSIONAL EXPERIENCE

Korea Advanced Institute of Science and Technology (KAIST)	Seoul, South Korea
Teaching Assistant	September 2019 – Present
Naver Cloud	Seongnam, South Korea
Research Intern	July 2023 – December 2023
– Developed an interactive image matting model using Segment Anything Model (SAM), with a focus on user-guided segmentation and boundary refinement.	

SCHOLARSHIP

KAIST Support Scholarship, KAIST	September 2019 – Present
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AWARDS

Compute Transparency Champion , IEEE/CVF Conference on Computer Vision and Pattern Recognition	2026
Best Poster Award , KAIST AI Workshop	2022
Silver Medalist , Korean Chemistry Olympiad	2006, 2007
1st Prize , The 1st Steinway & Sons Music Enthusiast Piano Competition	2022

TECHNICAL SKILLS

Languages: Python, MATLAB